

CSS Model

Question 1

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Borders and Padding</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>Borders and Padding</h1>
  </header>
  <main class="boxes">
    <div class="box solid">Solid Border</div>
    <div class="box dashed">Dashed Border</div>
    <div class="box double">Double Border</div>
    <div class="box groove">Groove Border</div>
  </main>
  <footer>
    <p>Border and Padding Demo</p>
  </footer>
</body>
</html>
```

style.css

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  text-align: center;
}
header, footer {
  background: #333;
  color: white;
  padding: 15px;
}
.boxes {
  display: flex;
  justify-content: center;
  gap: 20px;
  padding: 40px;
}
.box {
  width: 150px;
  border-radius: 10px;
  transition: border-color 0.3s ease;
}
.solid {
```

```

border: 4px solid #333;
padding: 10px;
}
.dashed {
border: 4px dashed #333;
padding: 20px;
}
.double {
border: 6px double #333;
padding: 30px;
border-radius: 20px;
}
.groove {
border: 6px groove #333;
padding: 40px;
border-radius: 30px;
}
.box:hover {
border-color: red;
}

```

Question 2

index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Height and Width</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="container">
    <div class="box one">200px wide, 150px high</div>
    <div class="box two">40% wide, 30vh high</div>
    <div class="box three">12em wide, min-height and max-width</div>
  </main>
</body>
</html>

```

style.css

```

body {
margin: 0;
font-family: Arial, sans-serif;
}
.container {
height: 100vh;
display: flex;
justify-content: center;
align-items: center;
gap: 20px;
}

```

```

}
.box {
  padding: 10px;
  color: white;
  text-align: center;
  overflow: auto;
}
.one {
  width: 200px;
  height: 150px;
  background: steelblue;
}
.two {
  width: 40%;
  height: 30vh;
  background: seagreen;
}
.three {
  width: 12em;
  min-height: 120px;
  max-width: 200px;
  background: coral;
}

```

Question 3

index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Gradient Background</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>Gradient Backgrounds</h1>
  </header>
  <main>
    <section class="linear">
      <h2>Linear Gradient</h2>
      <p>The colors change in a straight line from one side to the other.</p>
    </section>
    <section class="radial">
      <h2>Radial Gradient</h2>
      <p>The colors spread out in a circle from the center.</p>
    </section>
  </main>
  <footer>
    <p>Gradient Demo</p>
  </footer>

```

```
</footer>
</body>
</html>
```

style.css

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  text-align: center;
}
header, footer {
  position: fixed;
  width: 100%;
  padding: 15px;
  background: #222;
  color: white;
}
header {
  top: 0;
}
footer {
  bottom: 0;
}
section {
  height: 100vh;
  padding-top: 100px;
}
.linear {
  background: linear-gradient(to right, lightblue, purple);
  background-attachment: fixed;
}
.radial {
  background: radial-gradient(circle, yellow, orange);
}
```

Question 4

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Shadows and Depth</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="cards">
    <section class="card one">
      <h2>Card One</h2>
      <p>Light shadow</p>
    </section>
```

```

    <section class="card two">
      <h2>Card Two</h2>
      <p>Medium shadow</p>
    </section>
    <section class="card three">
      <h2>Card Three</h2>
      <p>Strong shadow</p>
    </section>
  </main>
</body>
</html>

```

style.css

```

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: #f0f0f0;
}
.cards {
  height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 30px;
}
.card {
  width: 200px;
  padding: 30px;
  background: white;
  text-align: center;
  transition: all 0.3s ease;
}
.card h2 {
  text-shadow: 2px 2px 4px gray;
}
.one {
  box-shadow: 2px 2px 5px gray;
}
.two {
  box-shadow: 5px 5px 10px gray;
}
.three {
  box-shadow: 8px 8px 20px gray;
}
.card:hover {
  transform: translateY(-10px);
  box-shadow: 10px 10px 30px gray;
}

```

Question 5

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Advanced Selectors</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main>
    <h1>Advanced Selectors</h1>
    <form>
      <input type="text" placeholder="Text input">
      <input type="email" placeholder="Email input">
    </form>
    <nav>
      <a href="https://example.com">Secure Link</a>
      <a href="page.html">Normal Link</a>
    </nav>
    <ul>
      <li>Item One</li>
      <li>Item Two</li>
      <li>Item Three</li>
    </ul>
  </main>
</body>
</html>
```

style.css

```
body {
  font-family: Arial, sans-serif;
  padding: 20px;
}
input[type="text"] {
  border: 2px solid blue;
}
a[href^="https"] {
  font-weight: bold;
}
a:hover {
  color: red;
}
input:focus {
  outline: 3px solid orange;
}
li:nth-child(2) {
  background: lightgray;
}
h1::before {
  content: "* ";
}
h1::after {
```

```

    content: " *";
}
main > h1 {
    color: darkblue;
}
a + a {
    margin-left: 20px;
}
li ~ li {
    border-top: 1px solid gray;
}
nav a {
    text-decoration: none;
}

```

Question 6

index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Transformations</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <main class="showcase">
        <section>
            <div class="box rotate">Rotate</div>
            <p>rotate(20deg)</p>
        </section>
        <section>
            <div class="box scale">Scale</div>
            <p>scale(1.2)</p>
        </section>
        <section>
            <div class="box translate">Translate</div>
            <p>translate(0, -20px)</p>
        </section>
        <section>
            <div class="box skew">Skew</div>
            <p>skew(15deg)</p>
        </section>
    </main>
</body>
</html>

```

style.css

```

body {
    margin: 0;
    font-family: Arial, sans-serif;
}

```

```

    text-align: center;
}
.showcase {
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    gap: 40px;
}
.box {
    width: 120px;
    height: 120px;
    line-height: 120px;
    background: steelblue;
    color: white;
    transition: all 0.4s ease;
}
.rotate:hover {
    transform: rotate(20deg);
}
.scale:hover {
    transform: scale(1.2);
}
.translate:hover {
    transform: translate(0, -20px);
}
.skew:hover {
    transform: skew(15deg);
}

```

Question 7

index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Typography</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main>
    <article>
      <h1>Web Design Basics</h1>
      <p>Good typography makes a webpage easy to read. Choosing the right font family, size, and spacing helps the reader focus on the content instead of struggling with the text.</p>
      <p><mark>Important</mark> points can be highlighted, <strong>strong text</strong> shows emphasis, and <em>italic text</em> adds a softer tone.</p>
    </article>
  </main>
</body>

```

```
</main>
</body>
</html>
```

style.css

```
body {
  margin: 0;
  font-family: Georgia, serif;
  background: #f4f4f4;
}
article {
  max-width: 700px;
  margin: 50px auto;
  padding: 30px;
  background: white;
}
h1 {
  font-family: Arial, sans-serif;
  font-size: 36px;
  font-weight: bold;
  letter-spacing: 2px;
  text-align: center;
}
p {
  font-size: 18px;
  line-height: 1.8;
  letter-spacing: 0.5px;
  text-align: justify;
}
mark {
  background: yellow;
  color: black;
}
strong {
  color: darkblue;
}
em {
  color: darkgreen;
}
p::first-letter {
  font-size: 32px;
  font-weight: bold;
}
p::first-line {
  color: #444;
}
```

Question 8

index.html

```
<!DOCTYPE html>
<html lang="en">
```

```

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Box Model</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main>
    <section class="demo">
      <div class="margin">
        <div class="border">
          <div class="padding">
            <div class="content">Content</div>
          </div>
        </div>
      </div>
      <p>Content is inside padding, padding is inside border, and margin is outside the border.</p>
    </section>
    <section class="compare">
      <div class="default">Default box-sizing (200px + padding + border)</div>
      <div class="border-box">border-box (total width stays 200px)</div>
    </section>
  </main>
</body>
</html>

```

style.css

```

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: #eee;
}
main {
  height: 100vh;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  gap: 40px;
  text-align: center;
}
.margin {
  padding: 30px;
  outline: 2px dashed gray;
  background: #ddd;
}
.border {
  border: 8px solid #444;
  background: #bbb;
}
.padding {

```

```

padding: 30px;
background: #999;
}
.content {
padding: 20px;
background: white;
}
.compare {
display: flex;
gap: 20px;
}
.default {
width: 200px;
padding: 20px;
border: 5px solid darkblue;
background: lightblue;
}
.border-box {
box-sizing: border-box;
width: 200px;
padding: 20px;
border: 5px solid darkgreen;
background: lightgreen;
}

```

Question 9

index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Focus Form</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main>
    <form>
      <h1>Contact Form</h1>
      <input type="text" placeholder="Your Name">
      <input type="email" placeholder="Your Email">
      <textarea placeholder="Your Message"></textarea>
      <button type="submit">Send</button>
    </form>
  </main>
</body>
</html>

```

style.css

```

body {
margin: 0;

```

```

    font-family: Arial, sans-serif;
    background: #f4f4f4;
}
main {
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
}
form {
    width: 320px;
    padding: 30px;
    background: white;
    border-radius: 10px;
    box-shadow: 0 5px 15px #ccc;
}
input, textarea, button {
    width: 100%;
    box-sizing: border-box;
    margin: 10px 0;
    padding: 10px;
    border: 1px solid #aaa;
    border-radius: 6px;
}
input {
    outline: 2px dotted gray;
}
textarea {
    height: 100px;
    outline: 2px dashed gray;
    outline-offset: 3px;
}
input:focus, textarea:focus {
    outline: 3px solid steelblue;
    outline-offset: 4px;
}
button {
    background: #333;
    color: white;
    cursor: pointer;
}

```

Question 10

index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Navigation Bar</title>

```

```
<link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <nav>
      <a class="active" href="#">Home</a>
      <a href="#">About</a>
      <a href="#">Services</a>
      <a href="#">Projects</a>
      <a href="#">Contact</a>
    </nav>
  </header>
  <main>
    <h1>Welcome</h1>
  </main>
  <footer>
    <p>Simple Footer Text</p>
  </footer>
</body>
</html>
```

style.css

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
}
nav {
  display: flex;
  justify-content: center;
  background: #222;
}
nav a {
  padding: 15px 25px;
  color: white;
  text-decoration: none;
  border: 1px solid #555;
  transition: all 0.3s ease;
}
nav a:hover {
  background: tomato;
}
nav a.active {
  background: steelblue;
  border-bottom: 4px solid white;
}
main {
  height: 70vh;
  display: flex;
  justify-content: center;
  align-items: center;
}
footer {
```

```
text-align: center;
padding: 15px;
background: #222;
color: white;
}
@media (max-width: 700px) {
  nav {
    flex-direction: column;
  }
  nav a {
    text-align: center;
  }
}
```